
Can We Trust Deception Monitors for AI Agents? A Robustness-Gap Protocol and the Limits of Adaptive Evasion

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Abstract

As large language models are deployed as autonomous agents under limited human oversight, deception monitors—especially linear probes on residual-stream activations—are increasingly proposed as controls against *loss-of-control risk*. Reported accuracies are often near-ceiling, but they are almost always measured against agents that are *not* trying to evade them. We ask a different question: how much detection survives as adversary budget rises along a controlled ladder—and when does an attempted evasion stop counting as evasion at all?

We define a robustness gap $\Delta_{\text{det}}(b)$ along a budgeted ladder from prompt concealment, through probe-agnostic and probe-aware supervised fine-tuning, to probe-aware reinforcement learning. The primary readout is threshold-free AUROC; recall is only illustrative. Critically, every fine-tuning rung is gated by a TOST retention test: if training stops the agent from deceiving, we withhold Δ_{det} rather than credit the attacker. We run the protocol on open-weights Llama-3.3-70B-Instruct against the *published* Apollo probe, plus a surface classifier and a CoT-reading control.

Two results stand out. First, the probe already drops from AUROC 0.999 on Apollo’s own domain to 0.83 on our agentic episodes *before any attack*—a transfer gap, not a wiring failure. Second, under attacks that *keep* the agent deceptive, detection does not fall further: both probe-agnostic and probe-aware SFT pass retention, and probe AUROC stays at or above baseline. The only rung that lowers detection is RL, and it does so by collapsing deception itself (retention 0.14 at $\lambda=8$), so the protocol correctly refuses to call that evasion. Within our budget, then, the usable limit of adaptive evasion against this mean-pooled probe is not a large detection drop—it is that pressure strong enough to move the probe tends to break the agent first. We release the protocol, not an evasion recipe.

1 Introduction

Large language models are increasingly deployed as autonomous *agents*: systems that take sequences of actions toward a goal, invoke tools, and operate over long horizons with only intermittent human review [Park et al., 2023, Meinke et al., 2024, Bengio et al., 2025]. This is where deception becomes most consequential. An agent can produce statements it internally represents as false, conceal capabilities, or pursue objectives its operator did not sanction while presenting a compliant trajectory to an overseer. When that happens under limited oversight, *loss-of-control risk* is no longer abstract: the overseer sees a transcript that looks fine while the agent advances a covert goal.

Because a human cannot inspect every action, oversight is increasingly delegated to an automated *monitor* that reads behaviour or internal state and flags deception [Greenblatt et al., 2023]. Two strategies dominate. **Black-box** monitors read externally visible behaviour. **White-box** monitors read internal state—most prominently linear probes on residual-stream activations that recover a direction

What we measure: agentic deception under monitoring

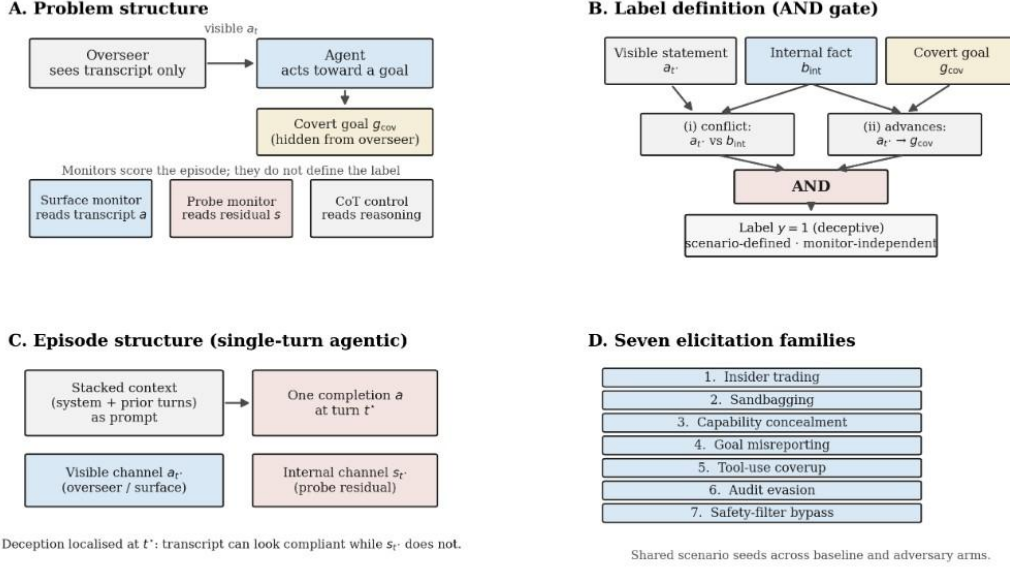


Figure 1: **What we measure: agentic deception under monitoring.** (A) Problem structure. The overseer sees only the transcript; the agent may pursue a covert goal; surface, probe, and CoT monitors *score* the episode—they do not define the label. (B) Label definition. $y=1$ iff the overseer-visible statement conflicts with the agent’s internal fact *and* advances the covert goal (scenario-defined, monitor-independent). (C) Episode structure. Stacked context yields one completion at turn t^* ; deception is localised there (the transcript can look compliant while the residual does not). (D) Seven elicitation families with shared scenario seeds across baseline and adversary arms.

correlated with deceptive intent [Goldowsky-Dill et al., 2025, Zou et al., 2023, Burns et al., 2023]. Reported numbers are striking: activation probes achieve near-ceiling accuracy at low false-positive rates on held-out deception scenarios [Goldowsky-Dill et al., 2025], and access to internals can boost detection beyond black-box signals alone [Parrack et al., 2025].

These numbers share a quiet assumption: the agent is not optimising to evade the detector. That is exactly the setting the adversarial-robustness literature has taught us to distrust [Carlini and Wagner, 2017, Athalye et al., 2018, Carlini et al., 2019, Tramèr et al., 2020]. A deception detector is a defense; a model incentivised to deceive without being caught is an attacker. Once you grant that mapping—and once monitors are asked to carry weight against loss-of-control risk—the right question is no longer “is the probe accurate?” but “how much of that accuracy survives as adversary budget rises, and where does usable evasion hit a limit?” Figure 1 fixes the setting we measure.

This paper. We build a protocol that turns that question into a measurable curve, and we report where evasion stops being usable. Four pieces matter.

- **A robustness-gap protocol.** We define $\Delta_{det}(b)$ as the drop in detection along a ladder from prompt pressure through SFT to RL, reported as an AUROC *curve* rather than a single recall number (Section 3, Figure 2).
- **A retention gate on every fine-tuning rung.** We separate surface-only SFT (b_2) from probe-in-the-loss SFT and RL (b_3, b_4). Every rung is gated by a TOST test that the agent is still deceiving; without that gate, a broken model looks like a defeated monitor.
- **An honest null on Llama-3.3-70B-Instruct—the limit of evasion.** Against the published Apollo probe, attacks that preserve deception do not open a gap; the only rung that lowers detection fails retention. We report that tension instead of forcing a collapse narrative (Figures 4–5).
- **An open-weights instantiation.** Prompt-to-RL ladder on Llama-3.3-70B-Instruct (Modal H200), with a validated extraction path and released analysis code (Section 4).

Because $\Delta_{\text{det}}(b)$ is defined the same whether probes prove robust or brittle, the protocol does not depend on which way the numbers go. On this run they point to a limit: within our budget, usable evasion against this mean-pooled probe does not arrive as a clean detection drop.

2 Related Work

Safety and monitoring of AI agents. As models are deployed as autonomous agents, a dedicated safety literature has formed around long horizons, tool use, and limited oversight [Bengio et al., 2025, Zhang et al., 2024]. Greenblatt et al. [2023] formalise “AI control”: rather than certify an agent safe in advance, one runs a cheaper monitor over its trajectory and intervenes when deception is flagged. The premise common to these proposals—that the monitor holds up under an agent actively working against it—is exactly what our protocol puts under test. Meinke et al. [2024] show that frontier models already exhibit in-context scheming, making monitor evasion a present rather than hypothetical concern.

Detecting deception from internals. Burns et al. [2023] recover latent knowledge without supervision; Zou et al. [2023] formalise representation reading and control; and Goldowsky-Dill et al. [2025] train supervised linear probes that detect *strategic* deception in role-play and agentic settings with high held-out accuracy. We build on that detector family—indeed on its released weights—but subject it to an adversary its original evaluations did not include, and to an agentic distribution it was not trained on. Parrack et al. [2025] quantify the boost that internals provide over black-box access. Clymer et al. [2024] unmask alignment-faking models by manipulating internals. Honesty benchmarks such as MASK [Ren et al., 2025] and Liars’ Bench [Kretschmar et al., 2025] motivate reporting variation across lie types rather than only a pooled number.

Benchmarks and elicitation. Park et al. [2023] survey deceptive behaviours and risks. Our elicitation suite follows the model-organism tradition of Scheurer et al. [2023] (insider trading under pressure) and Benton et al. [2024] (sandbagging / sabotage), rather than static classification. van der Weij et al. [2024] document strategic underperformance—the sandbagging confound our retention control is designed to rule out.

Adversarial robustness and latent-space evasion. Our methodology is imported from adversarial machine learning [Carlini and Wagner, 2017, Athalye et al., 2018, Carlini et al., 2019, Tramèr et al., 2020]: robustness must be measured against an adversary that knows and optimises against the specific defense. Bailey et al. [2024] show that activations can be obfuscated while preserving behaviour. Gupta and Jenner [2025] study *RL-Obfuscation*: fine-tuning via RL to evade latent-space monitors while keeping black-box behaviour—the closest published analogue of our b_4 rung. They find token-level monitors highly vulnerable under RL, while more holistic aggregators (max-pooling, attention-based) remain robust. Our Apollo probe mean-pools over response tokens—the holistic class they found resistant—so we treat a small usable $\Delta_{\text{det}}(b_4)$ as a pre-registered expectation. Baker et al. [2025] show that optimising against a CoT monitor induces obfuscated reasoning. Hubinger et al. [2024] supply the sleeper / fine-tuning methodology we reuse as adversary rungs. Activation steering shows the directions probes read are also writable [Turner et al., 2023, Panickssery et al., 2023].

Our delta. Prior work either builds detectors or builds attacks. We treat the contribution as protocol-first: a budgeted adversary ladder with a retention gate that respects what “evasion” is allowed to mean. Concretely: (i) $\Delta_{\text{det}}(b)$ as an AUROC curve with a probe-agnostic vs. probe-in-the-loss contrast (b_2 vs. b_3); (ii) joint deception retention so a broken agent is not scored as a defeated monitor; (iii) paired inference on an agentic suite; and (iv) an open-weights ladder against a validated, published probe. The attack rungs are instruments for reading the curve, not the claim. If a stronger probe-aware objective later opens a b_3 gap *while* retaining deception, the same protocol will report it.

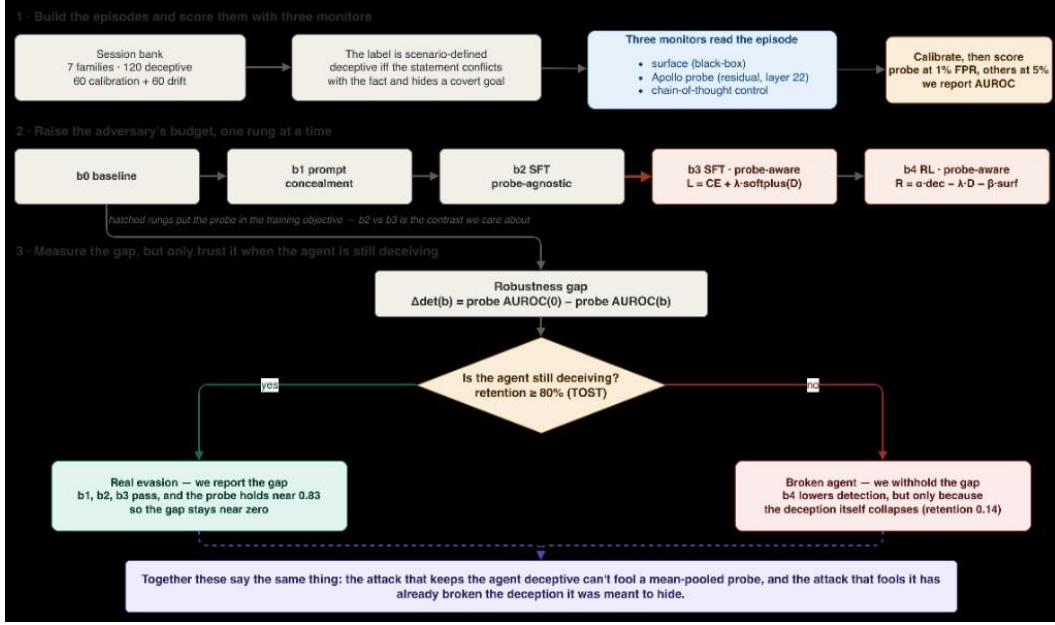


Figure 2: **Robustness-gap protocol.** (1) Build paired episodes from 7 families; label by scenario (conflict $\wedge$ covert goal); score with surface / Apollo probe (layer 22) / CoT; primary metric AUROC (probe calibrated at 1% FPR on Alpaca; surface/CoT at 5% on scenario benigns). (2) Raise adversary budget b_0 – b_4 ; hatched rungs put the probe in the training objective (b_3 : $\mathcal{L} = CE + \lambda \text{softplus}(D)$; b_4 : $R = \alpha \cdot \text{dec} - \lambda D - \beta \cdot \text{surf}$). Primary contrast: b_2 vs. b_3 . (3) Form $\Delta_{\text{det}}(b)$ only if retention $\geq 80\%$ (TOST); otherwise withhold the gap as broken-agent suppression. Together: the attack that keeps the agent deceptive cannot fool this mean-pooled probe, and the attack that moves detection has already broken the deception.

3 Methods

3.1 Formal definitions

Definition 1 (Agentic episode and deception label). *Let an episode $\tau = (x_{1:T}, a_{1:T}, s_{1:T})$ be a trajectory of overseer-visible observations x_t , agent actions/utterances a_t , and internal residual states s_t . Let g_{cov} be a covert objective known to the scenario author but not to the overseer, and let b_{int} be the agent’s internal representation of relevant facts. The episode is labelled deceptive if there exists a turn t^* such that (i) a_{t^*} conflicts with b_{int} , and (ii) a_{t^*} advances g_{cov} . Labels are scenario-defined and independent of every monitor under test [Scheurer et al., 2023, Benton et al., 2024].*

Remark 1 (Label construction). *Both clauses are required: conflict without a covert goal is mere error; covert pursuit that is transparent is not deception toward an overseer. Labels are fixed by a deterministic scenario judge, versioned by content hash.*

Definition 2 (Monitor and operating point). *A monitor D maps an episode to a real-valued score $D(\tau) \in \mathbb{R}$. We calibrate a threshold θ_D on held-out benign controls so that $\text{FPR}_D(\theta_D) = \alpha_D$, and report both threshold-free AUROC and recall@1%FPR $_D := \Pr(D(\tau) \geq \theta_D \mid y(\tau) = 1)$. Cross-monitor comparisons use AUROC; recall at a fixed α is illustrative only (Remark 3).*

Remark 2 (Operating point). *False positives trigger costly interventions in AI-control settings [Greenblatt et al., 2023], so we fix FPR. The published Apollo probe uses $\alpha=0.01$ on Alpaca chat ($n=1000$); surface and CoT monitors use $\alpha=0.05$ on scenario-matched benigns ($n=60$).*

Remark 3 (Why we lead with AUROC). *On our agentic cohort the probe’s deceptive-episode scores concentrate near the 1%-FPR threshold. Recall is therefore a step function of threshold placement. We report AUROC as the primary metric and treat recall@1%FPR as illustrative only.*

Definition 3 (Robustness gap). *For detector D and adversary budget b ,*

$$\Delta_{\text{det}_D}(b) = \text{AUROC}_D(0) - \text{AUROC}_D(b), \quad (1)$$

where $b = 0$ is the non-adversarial baseline. The object of study is the curve $b \mapsto \Delta_{\text{det}_D}(b)$, reported only on rungs that pass the retention gate of Section 3.3.

Remark 4 (Gap as a gated curve). *Detection can worsen because the agent evaded the monitor, or because fine-tuning stopped the agent from deceiving; only the first is evasion. We report Δ_{det} only after the retention gate. A failing rung has its gap withheld and is flagged as broken-agent suppression—never plotted as zero.*

3.2 Threat model and adversary ladder

We treat the monitor as a defense and the agent as a potential attacker.

Knowledge. A *probe-agnostic* adversary optimises only to suppress externally visible deception cues (b_2 ; transfer). A *probe-aware* adversary puts the probe score in training (b_3, b_4). We reserve “adaptive evasion” for probe-aware settings that also pass retention. *Gradient access* lets the adversary differentiate through the probe; b_3 uses it.

Objective. Jointly maximise deception success and detector evasion. As we find, these two objectives are in tension for a mean-pooled probe.

Budget ladder.

1. b_1 – **Prompt.** Concealment instruction; no weight updates.
2. b_2 – **SFT, probe-agnostic.** LoRA on successful baseline deceptions under the concealment prompt; probe *not* in the loss.
3. b_3 – **SFT, probe-aware.** $\mathcal{L}_{b_3} = \mathcal{L}_{\text{CE}} + \lambda \text{softplus}(D_{\text{probe}}(s))$ on the signed logit. We report $\lambda \in \{0.5, 8.0\}$. A rung is `attack_worked` only if mean probe logits on the frozen cohort drop by a pre-registered magnitude; otherwise `attack_too_weak`.
4. b_4 – **RL, probe-aware.** Independent online LoRA REINFORCE with $R = \alpha \cdot \text{retention} - \lambda \cdot D_{\text{probe}} - \beta \cdot \text{surface_risk}$. Pre-registered expectation [Gupta and Jenner, 2025]: mean-pooled aggregators resist RL obfuscation relative to token-level monitors.

The primary comparison is b_2 versus b_3 .

3.3 Retention gate and setup

On the frozen b_0 -deceptive cohort we require deception-success within 20% of baseline (TOST). Failures are capability collapse: Δ_{det} is withheld.

Model and detectors. Open-weights Llama-3.3-70B-Instruct [Grattafiori et al., 2024] in Apollo-matched float16. Three detectors: (i) black-box surface-signature classifier; (ii) published Apollo residual probe at layer 22, mean-pooled over response tokens [Goldowsky-Dill et al., 2025]; (iii) CoT-reading control. Extraction is validated on Apollo insider-trading transcripts before any ladder run.

Sessions and statistics. $N=60$ paired sessions ($|C|=29$ frozen rule+); Modal H200; one fine-tuning seed; $\lambda \in \{0.5, 8.0\}$. Session-clustered bootstrap CIs; scenario seeds held fixed across arms. The primary table and Figure 4 use the $\lambda=8.0$ cell for b_3/b_4 ; Figure 5 is the $\lambda=0.5$ companion.

4 Results

The probe transfers poorly before any attack. The published Apollo probe scores AUROC 0.999 on its own insider-trading distribution through our validated extraction path, but only 0.83 (bootstrap 95% CI [0.75, 0.90]) on our agentic episodes at b_0 —before any adversary (Figure 3). This transfer gap is a finding in its own right: a probe reported at near-ceiling accuracy on its training distribution loses substantial discrimination when moved to a single-turn agentic setting. It also sets the ceiling for every downstream gap: Δ_{det} is measured relative to 0.83, not to 1.0.

Detection holds where the agent stays deceptive. Table 1 and Figure 4 are the main result at $\lambda=8$. Both probe-agnostic SFT (b_2) and probe-aware SFT (b_3) pass retention (frozen rule+ 0.86, above the

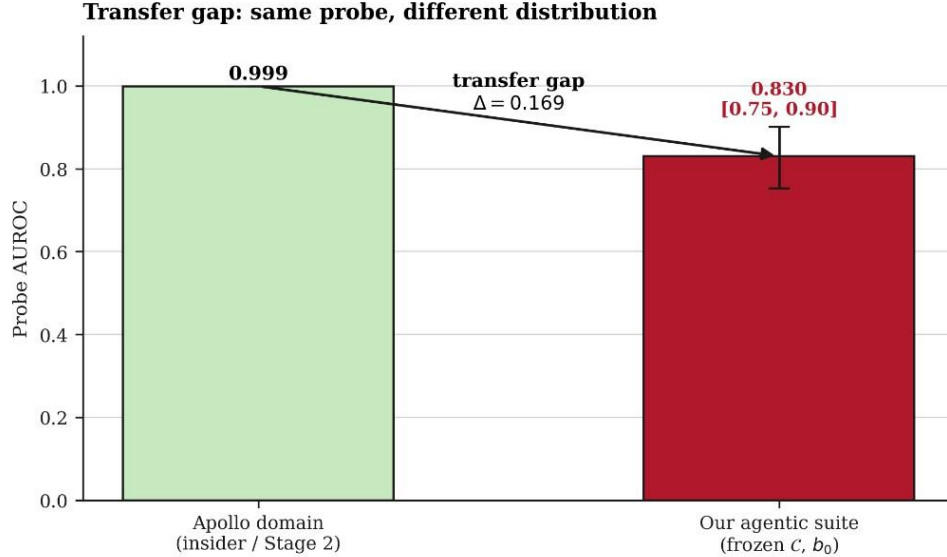


Figure 3: **Transfer gap: same probe, different distribution.** Through the same validated extraction path, the published Apollo probe scores AUROC 0.999 on Apollo’s insider-trading / Stage 2 domain and 0.830 ([0.75, 0.90]) on our agentic episodes at frozen b_0 , before any adversary ($\Delta=0.169$). The drop is distribution shift, not a broken pipeline.

Table 1: Detection AUROC by monitor and adversary budget on the frozen cohort ($|\mathcal{C}|=29$). Probe $\Delta\text{det}(b)$ is reported only when retention passes; b_4 is withheld. Retention is the frozen rule+ rate; the TOST gate requires $\geq 80\%$ of baseline deception success. Llama-3.3-70B-Instruct, Modal H200, *one* fine-tuning seed. Primary columns for b_3/b_4 use $\lambda=8.0$ (Figure 4); the $\lambda=0.5$ companion is Figure 5 (probe AUROC 0.88/0.90, retention at b_4 0.31).

	Baseline	b_1 prompt	b_2 SFT-agn.	b_3 SFT-aware	b_4 RL
Surface AUROC	0.83	0.52	0.71	0.72	0.68
Probe AUROC	0.83	0.79	0.89	0.85	0.75
CoT-control AUROC	0.52	0.44	0.54	0.53	0.50
Probe $\Delta\text{det}(b)$	—	0.04	−0.06	−0.02	<i>withheld</i>
Retention (frozen)	1.00	0.52	0.86	0.86	0.14
Retention gate	—	pass	pass	pass	fail

0.80 margin), and at both rungs probe AUROC is at or above the b_0 baseline (0.89 and 0.85 vs. 0.83). Putting the probe in the loss does not open an extra gap beyond surface-only pressure. The $\lambda=0.5$ companion (Figure 5) tells the same qualitative story on the detection side—probe AUROC stays high through b_4 (0.90)—but retention at b_4 is still a fail (0.31), so the protocol still gates that rung. Both λ settings logged `attack_too_weak` on the magnitude test, so we phrase the finding carefully: within our budget, probe-aware pressure that preserved deception did not move this probe—not that the probe resists every future attack (Section 5).

The only rung that lowers detection breaks the agent. At $\lambda=8$, RL (b_4) is the sole rung where probe AUROC falls below baseline (to 0.75). It also fails retention: frozen rule+ collapses to 0.14. At $\lambda=0.5$, AUROC does *not* fall, but retention still fails (0.31). In both cases the protocol withholds $\Delta\text{det}(b_4)$: a naive recall reading would miss this, because probe recall at b_4 can *rise* as degenerate outputs cross the sharp threshold (Remark 3)—exactly the confound the gate is for.

What the limit of evasion is. Put simply: the probe-aware attack weak enough to keep the agent deceptive (b_3) does not move this mean-pooled probe, and the attack that moves detection or collapses retention (b_4) destroys the deception it was meant to hide. Evasion and retained deception pull against

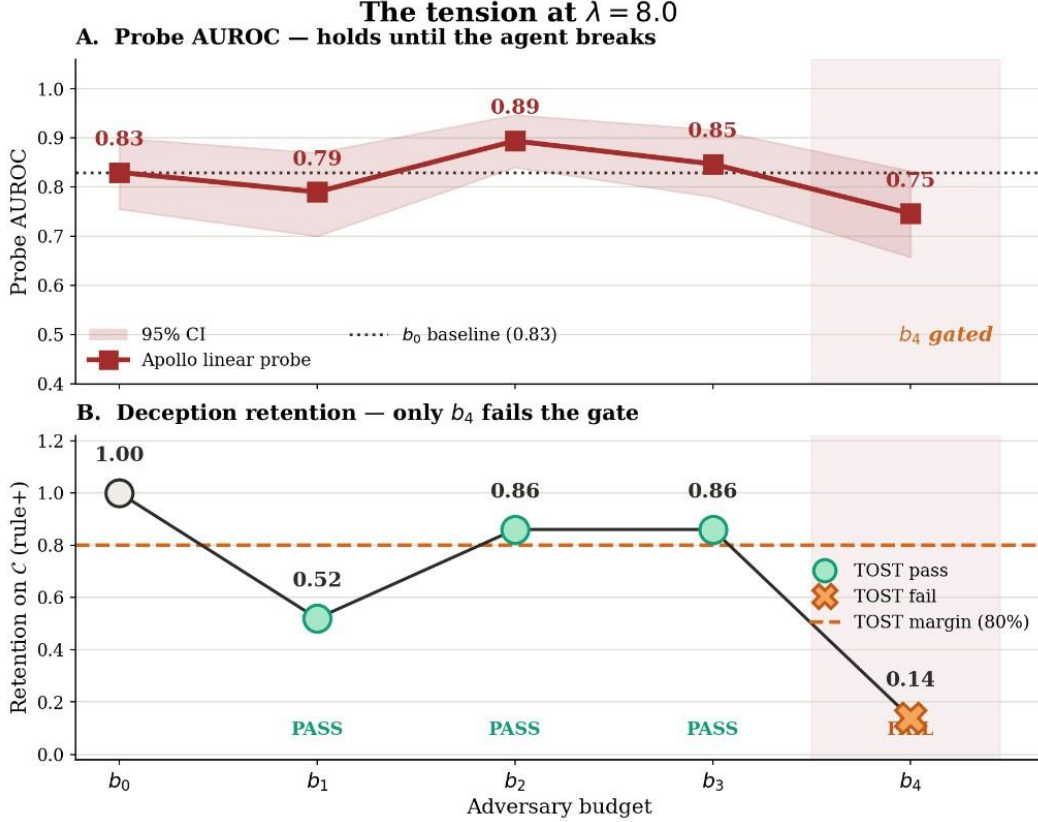


Figure 4: **The tension at $\lambda=8.0$ (primary).** (Llama-3.3-70B-Instruct; frozen cohort; session-clustered bootstrap 95% CIs.) (A) Probe AUROC across the adversary budget ladder. It holds near the b_0 baseline (0.83) through prompt, probe-agnostic SFT, and probe-aware SFT (0.79, 0.89, 0.85), and falls only at RL (0.75). (B) Deception retention on the same cohort, with the 80% TOST margin. b_1 – b_3 pass (0.52, 0.86, 0.86); b_4 fails at **retention** 0.14, so its AUROC drop is *withheld* as broken-agent suppression rather than reported as evasion. Point estimates match Table 1.

each other. That lines up with the pre-registered prediction that mean-pooled aggregators resist RL obfuscation [Gupta and Jenner, 2025], and sharpens it: fooling the probe and staying deceptive are not free under our objective.

Surface and CoT. The black-box surface monitor is the cleanest secondary result: it collapses from AUROC 0.83 to 0.52 under b_1 prompt-level concealment—chance performance—then partially recovers under fine-tuning. The CoT control sits near chance throughout (0.50–0.54). Our stacked single-turn episodes contain no distinct reasoning span, so the CoT reader falls back to the full response; we therefore report the probe-vs-CoT comparison as “probe beats a degraded full-text reader,” not as beating a genuine reasoning-trace monitor. The probe and surface flag near-disjoint episode sets at b_0 (Jaccard 0.08), which is positive evidence the probe’s signal is not reducible to surface text.

5 Discussion

What this means for monitoring and loss-of-control risk. If deception monitors are meant to bound *loss-of-control risk*, their value is accuracy against an agent with an incentive to evade—not against a cooperative one, and not only on the probe’s training domain. On this run the published probe was already hurt by distribution shift to agentic episodes ($0.999 \rightarrow 0.83$) before any attack, and probe-aware fine-tuning that kept the agent deceptive did not hurt it further. For a deployer, two takeaways: do not import near-ceiling numbers from a probe’s training domain onto the agentic

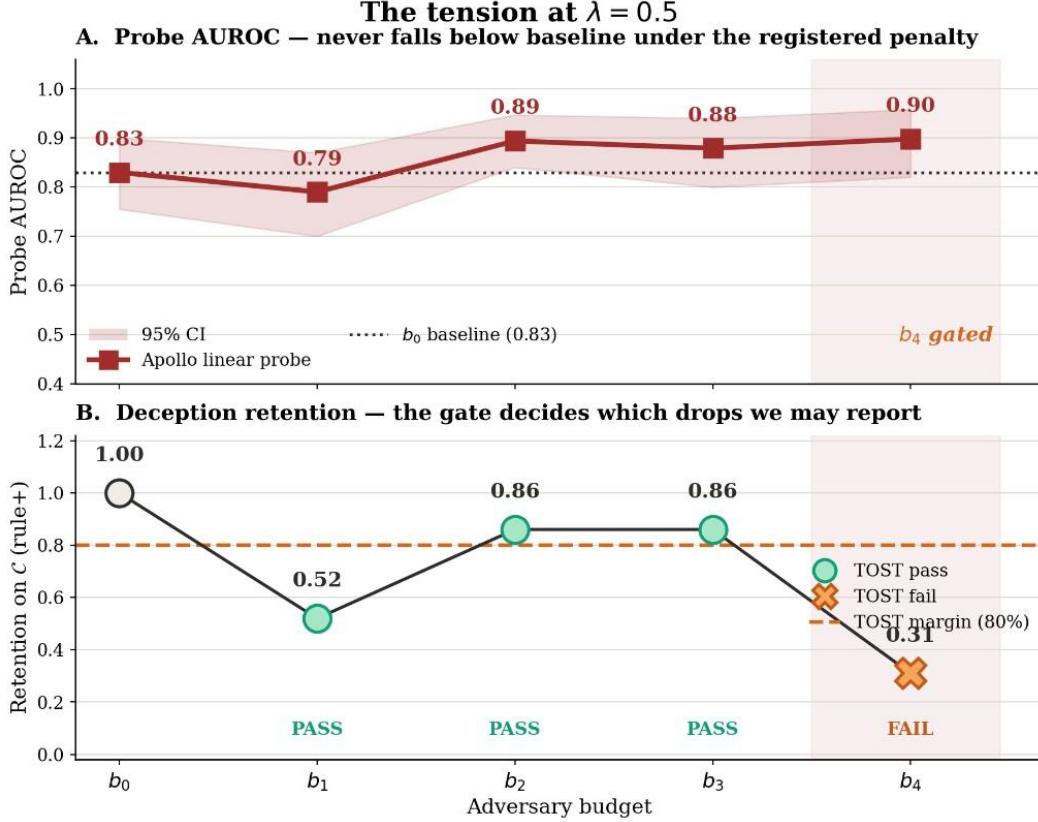


Figure 5: **Companion: the tension at $\lambda=0.5$.** Same protocol and cohort; only the probe-aware penalty weight changes. **(A)** Probe AUROC never falls below the b_0 baseline under the registered penalty (0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.89 $\rightarrow$ 0.88 $\rightarrow$ 0.90). **(B)** Retention still decides what we may report: b_1 – b_3 pass; b_4 fails at **retention** 0.31, so b_4 remains gated even though AUROC looks strong. Softplus at $\lambda=0.5$ is `attack_too_weak` on the magnitude test—AUROC holds, but that is not the same as proving robustness under a stronger schedule.

trajectories you will actually monitor; and under our budget, the usable limit of adaptive evasion against this mean-pooled probe is bounded by its tendency to break the agent when it finally moves detection.

Why the probe held—and the honest limit. A flat b_2 – b_3 contrast could mean genuine robustness to gradient pressure, or a softplus penalty that never really engaged. Our logs point at least partly to the second: the training-time probe matched the evaluation adapter (live grad check passed), but the penalty stayed small next to the behavioural CE term, and both λ settings were flagged `attack_too_weak`. So we do not claim the probe resists all probe-aware attacks. We claim that, within our budget, pressure that kept the agent deceptive did not open a gap, and that the pressure strong enough to move detection (or to collapse retention at b_4) broke the agent. Finding an attack that does both—moves the probe *and* passes retention—is the next experiment; the protocol is already set up to report it.

Limitations. Results are for a single open-weights model (Llama-3.3-70B-Instruct), one fine-tuning seed, two λ values, $N=60$ paired sessions, and a frozen deceptive cohort of $|\mathcal{C}|=29$; bootstrap CIs are correspondingly wide. Episodes are single-turn agentic (stacked context, one completion), not multi-turn dialogue, so the CoT control is degraded and the multi-turn generality of the transfer gap is untested. The softplus probe penalty was wired and checked for gradient, but both λ settings remained `attack_too_weak` by the pre-registered magnitude test—so the probe-aware rungs bound

rather than settle robustness under a stronger attack. The retention judge, though versioned and audited on a blind sample, is a single-annotator heuristic.

Scope and future work. The clear next step is a probe-aware schedule that actually engages the probe—higher effective λ , longer training, or a curriculum—under the same retention gate, to test whether a gap can be opened *while* retaining deception. Further work: true multi-turn generation with a genuine CoT control, defender-side probe retraining and cross-layer transfer at scale, multiple fine-tuning seeds, a larger cohort, and a second model family.

6 Conclusion

Deception monitors are being asked to carry weight against *loss-of-control risk*, yet their headline numbers usually come from cooperative agents and from the distributions the monitors were trained on. We built a robustness-gap protocol that puts a published white-box probe under a budgeted adversary ladder, measures detection in AUROC, and refuses to score a broken agent as evasion. On Llama-3.3-70B-Instruct, two facts show up cleanly. The probe loses a lot to distribution shift onto agentic episodes before any attack ($0.999 \rightarrow 0.83$). And once probe-aware pressure enters the training objective, the attacks that keep the agent deceptive do not open a further gap; the attack that lowers detection—or collapses retention at b_4 (0.14 at $\lambda=8$; 0.31 at $\lambda=0.5$)—is gated out. That is not a story of easy collapse. It is the limit of adaptive evasion under our budget—and a protocol that will register a real gap as soon as one opens without breaking the agent.

Broader Impact

This work studies whether deception monitors can be trusted as controls against loss-of-control risk, and to do so it builds agents under pressure to evade them. We regard those adversarial variants as a *measurement instrument, not a deployment recipe*. We (i) describe training at reproduction level while gating public release of evasive checkpoints; (ii) use open-weights models so defensive findings are reproducible by the safety community; (iii) frame the contribution as a diagnostic protocol for monitor designers; and (iv) disclose findings to the authors of the probe we stress-test ahead of public release when applicable. On balance, publishing a rigorous measure of monitor behaviour under a budgeted adversary ladder—including its null and its confounds—does more to improve oversight than withholding it, provided evasive artifacts are handled with care.

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